

Introduction to Computer Architecture

Sheet #3

Questions:

1. Explain why, in MARIE, the MAR is only 12 bits wide while the AC is 16 bits wide
2. What is the difference between RTL and RTN?
3. How does the CPU clock affect the RTL?

Exercises:

1. For the following program:
 - a) List the hexadecimal code.
 - b) What are the contents of the symbol table?

<u>Label</u>	<u>Hex Address</u>	<u>Instruction</u>
	100	Load A
	101	Add One
	102	Jump S1
S2,	103	Add One
	104	Store A
	105	Halt
S1,	106	Add A
	107	Jump S2
A,	108	HEX 0023
One,	109	HEX 0001

2. Given the following MARIE program, generate the symbol table from the first pass.

If,	100	Load	X	/Load the first value
	101	Subt	Y	/Subtract the value of Y, store result in AC
	102	Skipcond	400	/If AC=0 (X=Y), skip the next instruction /Note: This is Skipcond 01 in the book
	103	Jump	Else	/Jump to Else part if AC is not equal to 0
Then,	104	Load	X	/Reload X so it can be doubled
	105	Add	X	/Double X
	106	Store	X	/Store the new value
	107	Jump	Endif	/Skip over the false, or else, part to end of if
Else,	108	Load	Y	/Start the else part by loading Y
	109	Subt	X	/Subtract X from Y
	10A	Store	Y	/Store Y-X in Y
Endif,	10B	Halt		/Terminate program (it doesn't do much!)
X,	10C	Dec	12	/Assume these values for X and Y
Y,	10D	Dec	20	

3. Consider a program with the following requirements:

- Add the contents of memory location 940H to the contents of memory location 941H.
- Store the result (held in the accumulator register) back into memory location 941H.

Assume:

- The Program Counter (PC) holds the address 300H.
 - Memory location 940H contains the initial value 0003H.
 - Memory location 941H contains the initial value 0002H.
- a) Write the MARIE assembly program to perform the operations.
 - b) Provide your MARIE program's machine code (hexadecimal representation).
 - c) Complete the following fetch-decode-execute table for all the instructions.

Step	RTN	PC	IR	MAR	MBR	AC
Initial values		300	-	-	-	-
Fetch	$MAR \leftarrow PC$					
	$IR \leftarrow M[MAR]$					
	$PC \leftarrow PC + 1$					
Decode	$MAR \leftarrow IR[11-0]$					
	(Decode $IR[15-12]$)					
Get Operand	$MBR \leftarrow M[MAR]$					
Execute	$AC \leftarrow AC + MBR$					